

Figure 2: A year in the life of a kernel maintainer

The `make menuconfig` will launch a TUI-based user interface with the default configuration options shown in Figure 3.

In this tutorial, let us use the default configuration provided by the kernel. So select 'Save' and save the `config` with the file name `.config`. You can modify the options, but be sure about this, and understand the effects and consequences of changing them.

Step 5

Compile the main kernel by issuing the following command:

```
$ make
```

Or:

```
$ make -j4
```

The `-j` parameter helps to compile the kernel in parallel on four cores. This value can be set to any numerical value. However, setting it equal to the number of available cores gives the best compilation performance.

Step 6

In this step, we are ready to install the compiled kernel modules into our system:

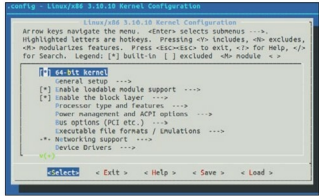


Figure 3: Options for kernel configuration

```
$ make modules_install
```

Step 7

Now, we are ready to install the kernel and boot into the system using the newly installed kernel. Issue the following commands on to the terminal:

```
$ make install
```

```
$ reboot
```

After rebooting, use the `uname -r` command to check the current kernel version. If you need to use a previous version of the kernel (or the pre-installed kernel), use the 'previously installed linux version' option from Grub.

The final word

Linux is a versatile kernel that is being used in various kinds of hardware architectures and supports thousands of devices. Thus, it is imperative to understand the manual configuration and compilation process of this kernel. **END**

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